# Arista Access Node WFD-ACC-71 Optical Transport Failure (SFP/XFP Degradation, Fiber Cuts, Rx/Tx Power Loss) Troubleshooting SOP

**Node:** WFD-ACC-71 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_8861

## Description

Optical transport failures on Arista Access nodes, specifically WFD-ACC-71 at the Whitefield POP, can occur due to a variety of reasons. The primary cause of these failures is often attributed to SFP/XFP degradation, which can be caused by overheating, aging, or physical damage to the transceivers. Additionally, fiber cuts or breaks can also lead to optical transport failures, resulting in Rx/Tx power loss. The underlying protocol mechanics responsible for this failure domain involve the Optical Transport Unit (OTU) and the Synchronous Digital Hierarchy (SDH) protocol, which are used for transporting data over optical fibers. The SDH protocol relies on the OTU to encapsulate and transmit data, and any issues with the OTU or SFP/XFP can disrupt this process, resulting in optical transport failures. Furthermore, the Arista Access node's role in the network, which involves providing Layer 2 switching and Layer 3 routing services, makes it critical to identify and resolve optical transport failures promptly to prevent network downtime and ensure seamless data transmission.

## Pre-requisite Checks

- Verify that the Whitefield POP is accessible and that remote hands are dispatched to the site.
- Check the status of the redundant link to ensure it is functioning properly.
- Gather relevant network and hardware logs to aid in troubleshooting.

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP Degradation | High |
| Fiber Cuts/Breaks | Medium |
| Rx/Tx Power Loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface status and SFP/XFP information, Description: This command displays the current status of all interfaces, including SFP/XFP information, which can aid in identifying issues with the SFP/XFP or fiber cuts.

Command: show inventory, Expected Output: Inventory of all hardware components, including SFP/XFP, Description: This command displays a detailed inventory of all hardware components, including SFP/XFPs, which can help identify any issues with the SFP/XFP or fiber cuts.
```

```
Command: show transceiver detail, Expected Output: Detailed SFP/XFP information, including temperature, voltage, and Rx/Tx power levels, Description: This command displays detailed information about the SFP/XFP, including temperature, voltage, and Rx/Tx power levels, which can aid in identifying issues with the SFP/XFP or fiber cuts.
```

## Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace-transceiver , Description: Replace the faulty SFP/XFP with a new one to resolve the optical transport failure.

Step 2: Step: Verify the fiber connections, Command: show interface status, Description: Verify that the fiber connections are secure and not damaged to prevent further optical transport failures.

Step 3: Step: Adjust the Rx/Tx power levels, Command: set transceiver power-adjust , Description: Adjust the Rx/Tx power levels to ensure optimal transmission and reception of data.

## Post-Check Verification

```
Command: show interface status, Expected Output: Interface status and SFP/XFP information, Description: Verify that the interface status and SFP/XFP information are normal after the resolution steps.

Command: show inventory, Expected Output: Inventory of all hardware components, including SFP/XFP, Description: Verify that the inventory of all hardware components, including SFP/XFPs, is updated and accurate after the resolution steps.

Command: show transceiver detail, Expected Output: Detailed SFP/XFP information, including temperature, voltage, and Rx/Tx power levels, Description: Verify that the detailed information about the SFP/XFP, including temperature, voltage, and Rx/Tx power levels, is normal after the resolution steps.
```

## Roll-back Procedures

- Step: Reinstall the original SFP/XFP, Command: install-transceiver , Description: Reinstall the original SFP/XFP in case the new one was not successful in resolving the optical transport failure.

- Step: Restore the default Rx/Tx power levels, Command: set transceiver power-adjust , Description: Restore the default Rx/Tx power levels in case the adjusted levels were not successful in resolving the optical transport failure.